

Effects of DL-3-n-Butylphthalide on Vascular Dementia and Angiogenesis

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Abstract 3-n-Butylphthalide (NBP) is a compound extracted from Chinese celery and is used as an anti-hypertensive herbal medicine for treating stroke patients. The aim of this study is to demonstrate the effects and mechanisms of this compound through in vitro and in vivo experiments. Culture experiments were performed by adding hydrogen peroxide (H_2O_2) to SH-SY5Y cells. From the MTT assay result, enhanced cell survival was observed with DL-NBP treatment, regardless of whether they are added before, simultaneously with or after the addition of H_2O_2 . For the in vivo experiment, Spontaneously Hypertensive rats and Wistar Kyoto control rats with chronic

cerebral ischemia, which were induced by bilateral transection of the common carotid arteries, were given DL-NBP. Their performances in the place navigation test and spatial probe test in the Morris Water Maze have significantly improved compared with the DL-NBP untreated animals, indicating an improvement in spatial learning and memory in the ischemic-animals. In addition, in the chick embryonic chorioallantoic membrane assay, angiogenesis was more vigorous under the effects of DL-NBP, together with increased expression of growth factors, VEGF, VEGF-receptor and bFGF. All these suggested that one of the mechanisms of DL-NBP might be ameliorating vascular dementia and promoting angiogenesis.

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Introduction

The brain, being one of the organs with greatest metabolic demand and yet possesses less antioxidants than other organs, is inevitably most susceptible to oxidative damage [1]. Neurodegenerative disorders, such as vascular dementia will lead to a gradual loss of memory, judgment, spatial orientation as well as other functional deficits of motor and sensory activities. Patients of vascular dementia are usually above 65 years old [2], and are shown to be more prevalent in Asia [3]. Despite the advance in medical technologies, vascular dementia is still a major problem of concern as statistics have shown a doubling in number of patients every 5.3 years [2]. The two main causes of vascular dementia are cerebral ischemia and oxidative stress. Oxidative stress is featured by the disruption of oxidative balance in our body, thus having insufficient antioxidants to overcome the increasing amount

of reactive oxygen species (ROS) produced during normal metabolic processes. ROS are highly unstable compounds that may or may not contain an unpaired electron, for example superoxide anion ($O_2^{\cdot-}$), hydroxyl radicals ($\cdot OH$) and hydrogen peroxide (H_2O_2) [4]. While these chemicals are essential in signaling pathways and have important physiological roles [1], the imbalance of them can cause direct vascular damage [5]. Oxidative stress can be both the cause and the effect of cerebral ischemia [6, 7], which eventually leads to the reduction of proteins that are essential for learning and memory [8], thus resulting in dementia.

DL-3-n-Butylphthalide (NBP) is a racemic mixture of optical isomers of compound extracted from seeds of *Apium graveolens* Linn, also known as Chinese celery. Regardless of the different isomers, many researches regarding NBP had shown promising results on its beneficial effects, both protective and preventive, in various disorders of the cerebral models, such as Parkinson's disease, stroke and brain edema [9–11]. For example, after administration of this compound into rats which had occlusion of the middle cerebral artery, the brain edema resulted from the occlusion was reduced [9]. NBP has also been used as an antihypertensive herbal medicine in human, and has been suggested to reduce neuronal apoptosis [12], increase cerebral microcirculation and energy metabolism, as well as reducing inflammation and oxidation [13, 14].

Although NBP has entered into phases of clinical trials in China [13], basic research was scanty [9–14], a better understanding of the additional functions and mechanism of actions of this compound would help us to project on the future pharmacological or nutraceutical use of this drug in neurodegenerative disorders. Using Evan blue as a marker, Chong and Feng [15] noted that NBP could increase blood flow after subarachnoid damage in rats. Utilizing the middle cerebral occlusion model in the rats, Li et al. [16] reported an up-regulation of VEGF and bFGF (both protein and RNA) in the peripheral infarct but not in the core infarct of rat brain. The question is therefore, does the drug NBP really enhance angiogenesis.

In this study, the effect of NBP on angiogenesis was evaluated using chick embryonic chorioallantoic membrane (CAM) culture as a model, in order to assess possible induction of more blood vessels. At the same time, the neuronal effects of NBP were tested using both in vivo and in vitro models. In order to mimic the conditions for vascular dementia in animals, Spontaneously Hypertensive rats (SHR) and Wistar Kyoto (WKY) control rats that had bilaterally ligated common carotid arteries were used. NBP was then administered to these hypertensive and normotensive rats with cerebral ischemia, and the neuronal impact of NBP in these rats was tested through the performance of Morris Water Maze, a behavioral test widely

used to examine the spatial learning and memory of the rats [17], abilities which are gradually lost in dementia sufferers. In addition, to test the effects of NBP on neuronal cells which had experienced oxidative stress, H_2O_2 were added to the SH-SY5Y cell culture in the presence or absence of NBP in the in vitro experiment. The hypothesized neuroprotective effects of NBP in terms of cell survival were then evaluated using the MTT assay.

Materials and Methods

Chorioallantoic Culture

Drugs and Reagents

Fertilized specific pathogen-free chicken eggs were purchased from Tai Lung Experimental Station (New Territories, Hong Kong). DL-3-n-Butylphthalide (purity, 99.6%; lot number, 09100151) was obtained from Shijiazhuang Pharma Group NBP Pharmaceutical Co., Ltd (Shijiazhuang, Hebei, China). Polyclonal Rabbit anti-VEGF antibody (Cat. No. 18-0254), normal goat serum (Cat. No. PCN5000), Biotin-Goat Anti-Rabbit IgG (H + L) (Cat. No. 65-6140) and HRP-Streptavidin Conjugate (Cat. No. 43-4323) were bought from Invitrogen Corporation (Camarillo, CA, USA). VEGF Receptor 2 (55B11) Rabbit mAb was purchased from Cell Signaling Technology, Inc. (#2479; Beverly, MA, USA). bFGF antibody was purchased from Santa Cruz Biotechnology, Inc. (sc-79; Santa Cruz, CA, USA). Other chemicals were obtained from Sigma-Aldrich Co. LLC. (St. Louis, MO, USA).

In order to transfer DL-NBP, qualitative 6 mm diameter filter paper pieces were punched out and were sterilized by high temperature and high pressure. They were then immersed in equal volumes of DL-3-n-Butylphthalide solution with different concentrations (0.1, 1.0, 10 μM NBP) for 5 min before use. Control group filter papers were treated by an equal volume of normal saline.

Chick Embryo Chorioallantoic Membrane (CAM) Assay

The CAM experimental protocol was taken from a previous procedure reported by Kirchner et al. [18] with minor modifications. Briefly, fresh fertile chicken eggs were cleaned, weighed to the nearest microgram, and 40 nearly-equal-weighted eggs were randomly divided into 4 groups, with 10 eggs in each group. These groups included the NBP 0.1 μM group, NBP 1.0 μM group, NBP 10 μM group and the control group. After grouping, the eggs' shell were sterilized by 75% alcohol, labelled and transferred into a hatching incubator with a relative air humidity of 65% and

a temperature of 37°C. The eggs were then placed according to the incubator manufacturer's instruction and rotated along their long axis every 3 h to prevent adherence. On day 7, a hole of approximately 1.5 cm × 1.5 cm was opened in the eggs' shell by surgical blade, cleared away with an ophthalmic forceps and exposing the shell membrane. Then, another hole in the shell at the air pouch of the egg was made, air was aspirated from this blunt end of the egg, so that the shell membrane was separated from the CAM. The CAM was then fully exposed for better access. For the regional application of NBP, NBP-soaked sterilized filter papers of different concentrations (0.1, 1.0, 10 µm NBP as well as control with saline) were deposited on the CAM of the different groups of eggs. All holes were then closed with a disinfected Laboratory Wrapping Film to prevent dehydration or possible infections. The position of depositing each filter paper in each egg was defined between two corresponding large blood vessels, which could be visualized and marked on the shell after observing with a light box. The eggs were then returned to the incubator in a static position for 7 more days. After incubation, the film on the egg was discarded and survival of the eggs was confirmed. Only the survived eggs were used in the further experiments, where fixation was done by adding 4% paraformaldehyde (PFA) for 2 h, and the CAM with the carrier was carefully separated from the embryo, removed, and placed on the cell culture dish after washing with phosphate buffered saline (PBS). After slight drying, the CAM vasculature was captured with a digital camera and images were analyzed with Image-pro plus 6.0 software (Media Cybernetics, Inc., Bethesda, MD, USA). To calculate the mean CAM blood vessel proportion, a round area of 1 cm radius centering around the filter paper was defined. The CAM blood vessel area for each sample with different concentrations of NBP was then digitized by the software according to the color versus gray values, and the ratio of the area of blood vessel to total area was calculated.

Immunohistochemistry

To detect the expression of VEGF, VEGF Receptor 2 and bFGF, immunohistochemistry were performed. The CAMs were fixed by 4% PFA for 2 h, dehydrated and embedded in wax. Then, a 5 µm sections were cut, dewaxed, and rehydrated. Antigen retrieval was performed with 0.01 M Citric buffer (pH 6.0) in a microwave at 850 W for 15 min. The sections were cooled down to room temperature, and endogenous peroxidase activity was blocked with 3% H₂O₂ in methanol for 10 min. Non-specific antigen was then blocked with 5% normal goat serum for 1 h, and subsequently incubated at 4°C overnight with primary antibodies against VEGF (1:200), VEGF Receptor 2 (1:600) and bFGF (1:300) diluted with 2% normal goat serum. After a thorough

wash with PBST (PBS containing 0.05% Tween-20), sections were incubated with Biotin-Goat Anti-Rabbit IgG (H + L) (1:500) in diluting solution for 1.5 h and then with HRP-Streptavidin Conjugate (1:400) in PBS for 1.5 h. The sections were then washed in PBST, and developed in PBS solution that contains 0.05% 3',3'-diaminobenzidine and 0.01% H₂O₂, in order to visualize the peroxidase bound to the complex. After color development, the sections were dehydrated through graded series of alcohol, cleared in xylene and mounted in permount. Control group was performed using 2% normal goat serum without the primary antibody. Finally, all images were photographed with an Zeiss Axiophot 2 microscope (Carl Zeiss MicroImaging, LLC, Thornwood, NY, USA) and equipped with a Spot RTKE CCD microscope camera using Spot Advanced software (Diagnostic Instruments Inc., Sterling Heights, MI, USA).

Statistical Analysis

All data from the morphometric study on the density of blood vessels per total area were reported as mean ± standard deviation (SD). To confirm the significant difference between groups, one-way ANOVA analysis was performed, followed by a post hoc comparison test using Fisher's LSD test with the software SPSS16.0 (SPSS Inc, Chicago, IL, USA). $P < 0.05$ was considered statistically significant.

Animal Experiment

Animal Experiment

Thirty-six 3-month-old male SHR and WKY rats (half SHR and half WKY) were used in this experiment with the approval of the Animal Experimentation Ethics Committee of the Chinese University of Hong Kong. The rats were evenly divided into three groups, with twelve rats per group. The three groups were: NBP-treated group with chronic cerebral ischemia, chronic cerebral ischemia group only and the sham-operated group.

Bilateral occlusion surgery was performed on both the NBP-treated group and the chronic cerebral ischemia group. Based on previous publications [19, 20], the model of bilateral occlusion with minor modifications was performed. In order to avoid incomplete occlusion of the bilateral common carotid artery (CCA) and reestablish of blood flow, the bilateral CCA was cut instead of ligated.

Sterilized equipments and tools were used in the operation. A mixture of ketamine (75 mg/kg) and xylazine (10 mg/kg) (both from Alfasan Group Company, Woerden, Holland) was injected intra-peritoneally. Then, a 1.5–2 cm cut was made in the midline of the disinfected neck.

The CCA was carefully isolated to prevent injury to the nerve fibers. Afterwards, the distal and proximal parts of the CCA were ligated with 5-0 sterile polypropylene suture (Somerville, NJ, USA), producing a 0.5 cm gap. The CCA was then transected between the two ends, and this induced cerebral chronic hypoperfusion. The wound was sutured and amoxicillin (Alfasan Group Company, Woerden, Holland) was applied locally to prevent infection. Analgesic Buprenorphine (0.01–0.05 mg/kg s/c, 2 times per day) was also given for 3 days after surgery.

The sham-operated group received a surgery without touching the vessel. After surgery, NBP dissolved in vegetable oil was given intragastrically to the NBP-treated group once per day, at a dose of 60 mg/kg. Same volume of vegetable oil without NBP was given daily to the other two groups. After 4 months of administration, the place navigation test and spatial probe test were performed in the Morris Water Maze to investigate the spatial learning and memory of the rats.

In order to assess the spatial learning and memory of the rats, we followed the protocol of Morris [21] while making slight modifications. The water maze was a circular pool with 100 cm diameter, 60 cm high, filled with water up to 42 cm, with the temperature of $(25 \pm 1)^\circ\text{C}$, and divided into four quadrants. A circular platform with diameter of 10 cm and height of 40 cm was placed 2 cm underwater in one of the four quadrants. The interior surface of the pool and the platform were black, and the environment was kept constant. On the day before experiment, rats were put in the pool without platform for 120 s of adaptive swimming.

Place Navigation Test

The test was performed twice per day for 4 days. In this experiment, each rat was released from one of the four quadrants of the pool facing the interior wall, letting the rat to seek for the platform. The time spent for each rat from being released to reaching the platform was recorded. This is the escape latency. If the rat found the platform, it would be kept there for 10 s, then returned to its home cage. After resting for 60 s, the rat was placed into another quadrant of the pool. If the rat failed to find the platform in 120 s, it would be led to the platform and kept standing for 10 s. In this case, the potential latency will be 120 s. The average escape latency of the last test was recorded and used for calculation of the average value of each group for comparison.

Spatial Probe Test

On the fifth day of the maze experiment, the platform was removed and each rat was released from the quadrant that

is opposite to the designated quadrant. The designated quadrant is the quadrant that had been containing the platform. The time spent by the rat to find and enter the designated quadrant was recorded.

Cell Culture Experiment

Neuronal Culture Experiment

Dulbecco's modified Eagle's medium (DMEM), F12K, fetal bovine serum (FBS), 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), penicillin and streptomycin were purchased from GIBCO. Dimethyl sulfoxide (DMSO) was bought from Sigma-Aldrich (St. Louis, MO, USA). Human neuroblastoma SH-SY5Y cell line was purchased from American Type Culture Collection (ATCC) (Rockville, MD, USA). Human neuroblastoma SH-SY5Y cells were cultured in 1:1 mixed DMEM and Ham's F12 medium with 10% FBS, 100 U/ml penicillin and 100 mg/ml streptomycin at 37°C and in the culture incubator with 5% CO_2 .

1×10^4 cells were seeded into each well of the 96-well plate. After 12 h, cells were attached to the plate. The medium was carefully removed and cells were treated in the following ways:

- Medium with different concentrations of NBP, NBP 1.0 μM and NBP 10 μM , was added, and the cells were cultured for 24 h. NBP was first dissolved in DMSO and then diluted by medium of different concentrations. The final concentration of DMSO in medium was less than 0.2% (v/v).
- Medium with different concentrations of NBP, NBP 0.1 μM and NBP 1.0 μM , was added and cells were cultured for 24 h. The medium was then removed, and replaced with a medium of 150 μM H_2O_2 . Cells were then cultured for another 12 h.
- Medium with different concentrations of NBP, NBP 0.1 μM and NBP 1.0 μM , was added simultaneously with 150 μM H_2O_2 . Cells were then cultured for 24 h.
- Medium with 150 μM H_2O_2 was added. After 12 h, the medium was replaced by different concentration of NBP, NBP 0.1 μM and NBP 1.0 μM . Cells were then cultured for another 24 h.

MTT Assay

In order to observe the viability of SH-SY5Y cells under the stress of H_2O_2 , the MTT assay was performed as previously described while making minor modifications [22]. Firstly, 10 μl 5 mg/ml of MTT was added into each well, and cells were incubated in the incubator of 37°C for 4 h. Then, the medium was carefully removed and 200 μl of

DMSO was added. The plate was shaken in dark for 10 min. Lastly, 490 nm wavelength absorbance was measured by Victor-2 Multilabel counter (PerkinElmer Life And Analytical Sciences, Inc., Waltham, MA, USA).

Statistical Analysis

All data were presented as mean $\pm$ SD. Statistical analysis was performed by one-way ANOVA followed by a post hoc comparison test using Fisher's LSD test to confirm the significant difference between the groups. $P < 0.05$ was considered statistically significant. All statistical procedures were performed with the software SPSS16.0 (SPSS Inc, Chicago, IL, USA).

Results

Chorioallantoic Culture

Compare Angiogenesis Between NBP Treatment Group and Saline Treatment Group

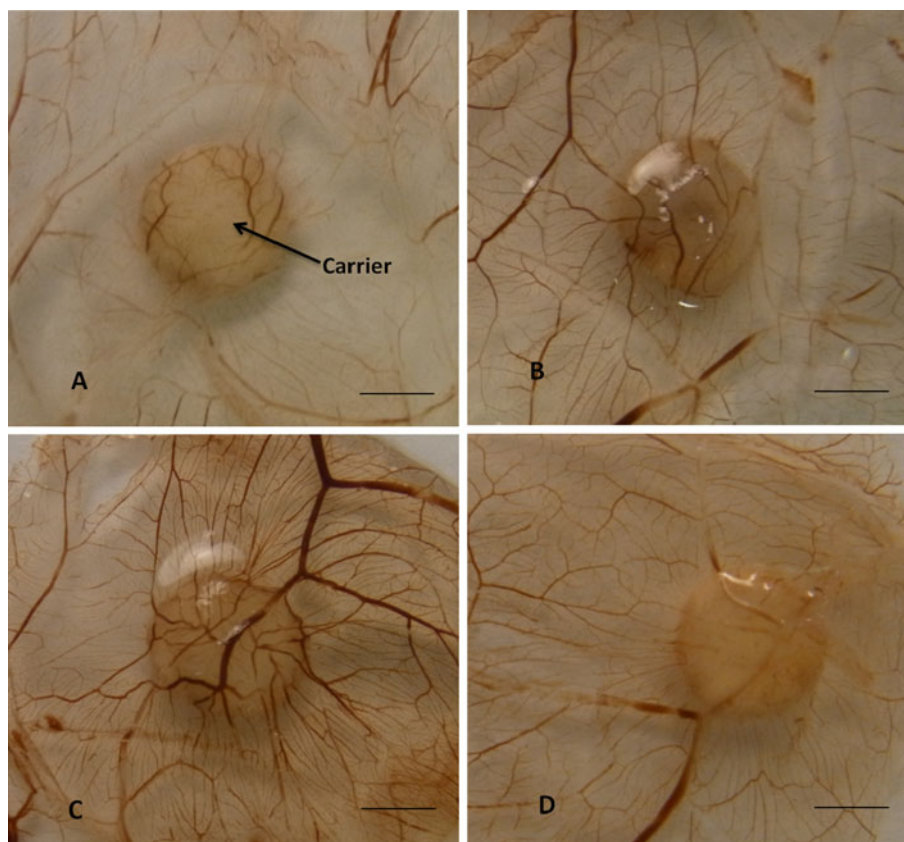
The areas surrounding the carrier were selected for evaluation of the growth of blood vessels in the CAMs. After

treating with different concentrations of NBP for 7 days, the number of blood vessels was compared between the NBP group and the saline group. These blood vessels were small in sizes and could possibly be newly formed vessels. Compared with the NBP group, the saline group did not support the growth of blood vessels, and was presented with few and scattered number of blood vessels (Fig. 1a). However, the NBP-treated groups had significantly increased number of blood vessels. Among them, those treated with higher NBP concentration had more blood vessel growth than those treated with lower concentration (Fig. 1b–d). Quantitative evaluation showed that NBP treated groups had indeed higher densities of blood vessels compared with the saline group (Fig. 2). However, when the NBP dosage reached 1.0 μm , the optimum effect was established (Fig. 2).

Characteristics of Potential Angiogenic Factors Involved in Blood Vessel Ingrowth

To illustrate the soluble growth factors which might be involved in NBP-induced angiogenesis, immunohistochemistry using antibodies against two potent angiogenic factors, VEGF and bFGF, was performed. We found that the distribution of the VEGF-, VEGF Receptor 2 and

Fig. 1 Blood vessels ingrowth surrounding the carrier. **a** Is the control group, **b–d** represents the 0.1, 1.0 and 10 μm NBP treatment group respectively. Bar 3 mm



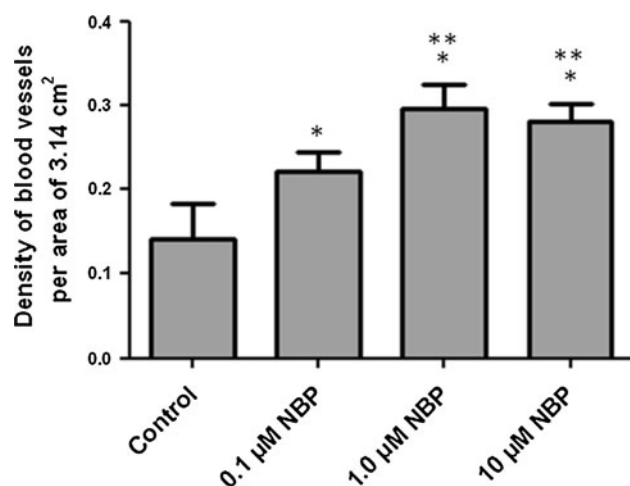


Fig. 2 Effects of NBP on blood vessel density determined by the CAM assay. The ability of angiogenesis induced by NBP was determined by the ratio of the blood vessel area to the selected total area for different concentration. Data were represented as averages of density per unit area $\pm$ SD. Data represented four separate experiments, each of forty eggs in each of the 4 groups (0.141 ± 0.042 , 0.222 ± 0.023 , 0.297 ± 0.029 , 0.282 ± 0.020 , respectively). * $P < 0.01$ versus control group; ** $P < 0.05$ versus 0.1 μ M group

bFGF-positive cells were different between the NBP groups and the control group after NBP treatment was added on these membranes. After 0.1 μ M NBP treatment, the expression levels of VEGF, VEGF Receptor and bFGF were more intensive on both the external and internal surface of the CAM and in the layers of the blood vessels compared to that of the control (Fig. 3). Greater increase in intensity of reaction was observed in those treated with higher concentrations (1.0 and 10 μ M). A semi-quantitation between groups was illustrated in Table 1.

Animal Experiments

After bilateral sectioning of the common carotid arteries on the rats, 80% of the rats survived. It appeared that the

Table 1 Semiquantitation on the relative optical densities of the chorioallantoic membranes after immunocytochemical reactions

	VEGF	VEGF receptor	bFGF
Control	+	+	+
0.1 μ M NBP	++	+	+
1 μ M NBP	++++	+++	+++
10 μ M NBP	+++	+++	+++

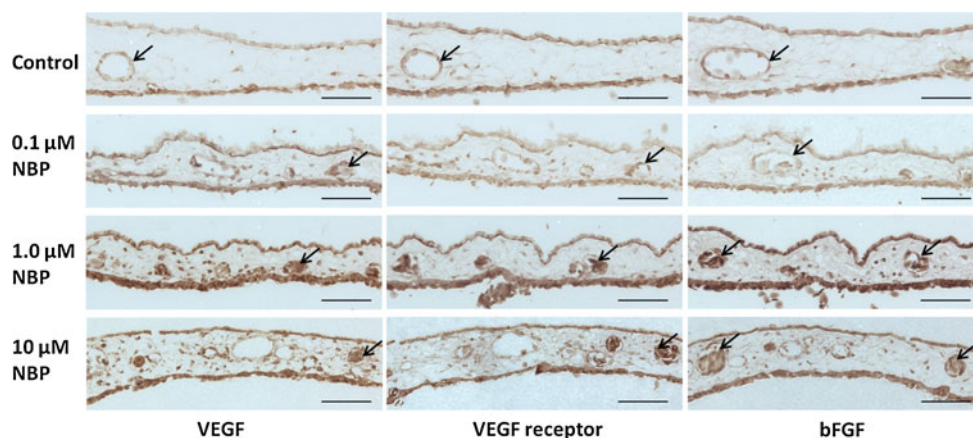
More “+” indicates a higher intensity of reaction

vertebral arteries in these survived rats, that had maintained the perfusion via the circle of Willis, could sustain the circulation of the brains.

Morris Water Maze

The effects of NBP on escape latency and the percentage time of swimming in supposed quadrants of SHR and WKY rats: (1) the place navigation test indicated that, 4 months after the SHR and WKY rats had experienced chronic cerebral ischemia, the escape latency of them increased significantly when compared to the sham-operated group. Also, the escape latency of NBP-treated ischemia group decreased significantly when compared to the untreated chronic cerebral ischemia group ($P < 0.01$, Fig. 4). (2) The spatial probe test indicated that, after the platform was removed, the traces of the rats in the different groups were different. The rats in the sham-operated group could find the platform with purpose and ease, and moved to the supposed quadrant directly with a strong tendency to stay inside the quadrant. In contrast, the untreated chronic cerebral ischemic rats moved without purpose, with random traces and with a tendency of swimming along the margin of the pool. The rats in the NBP-treated ischemic group moved in a pattern between the sham-operated group and the untreated chronic cerebral ischemic group. Numerically, the ratio of tP/tT swimming time in supposed

Fig. 3 Immunostaining of VEGF, VEGFR and bFGF on CAM after NBP or saline treatment. The darkly stained particle is the positive cell. Arrow on some blood vessels. Bar 50 μ m



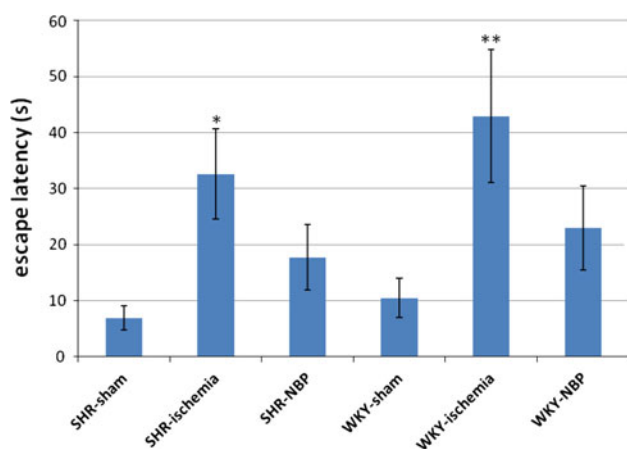


Fig. 4 The effects of NBP on the escape latency of SHR and WKY rats with chronic cerebral ischemia. Compared with SHR-sham group and SHR-NBP group, * $P < 0.01$; Compared with WKY-sham group and WKY-NBP group, ** $P < 0.01$

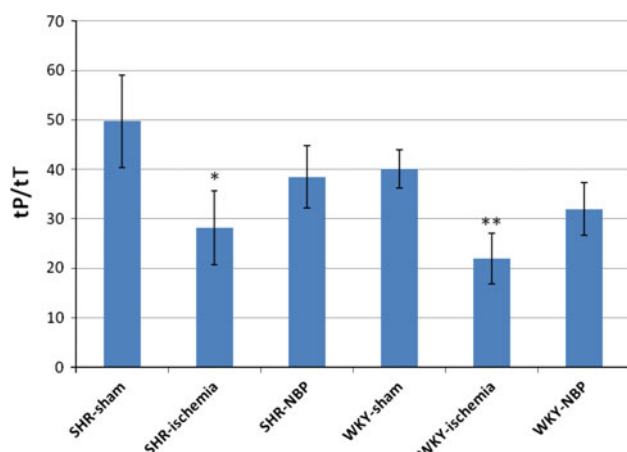


Fig. 5 The effects of NBP on the percentage time of swimming inside the supposed quadrant of SHR and WKY rats with chronic cerebral ischemia. Compared with SHR-sham group and SHR-NBP group, * $P < 0.01$; Compared with WKY-sham group and WKY-NBP group, ** $P < 0.01$

quadrant to total time (120 s) had significantly decreased in the chronic cerebral ischemic group compared to the sham-operated group, while the ratio in the NBP-treated group increased significantly compared to the chronic cerebral ischemic group ($P < 0.01$, Fig. 5). When these results were compared between the normotensive WKY and the hypertensive SHR rats, NBP was shown to protect both species at ischemic conditions (Figs. 4, 5).

Cell Viability in Cell Culture Experiment

The SH-SY5Y cells were treated in 4 different ways based on the time. NBP was added to the culture. Two different concentrations of NBP, 1.0 and 10 μM NBP, were added either alone, before, simultaneously with or after H_2O_2 was

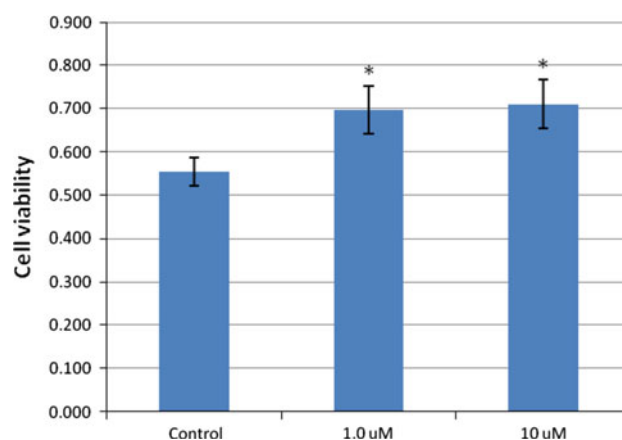


Fig. 6 The effects of the different concentrations of NBP to the viability of SH-SY5Y. Compared with control, * $P < 0.05$

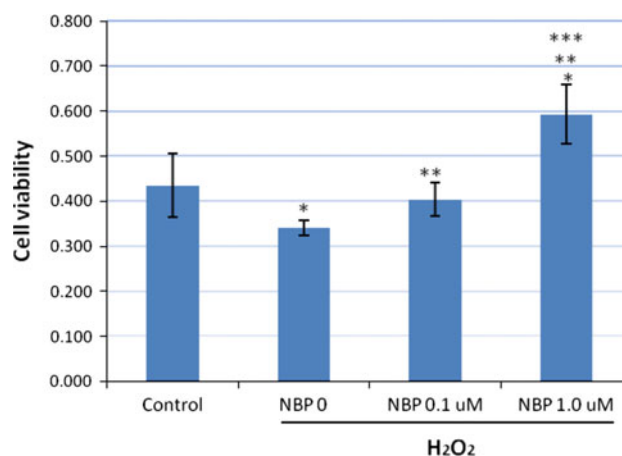


Fig. 7 The effects of the different concentrations of NBP to H_2O_2 pretreatment caused SH-SY5Y cell damage. Compared with control, * $P < 0.01$; Compared with NBP 0, ** $P < 0.01$; Compared with NBP 0.1 μM , *** $P < 0.01$

added. The results showed that with no H_2O_2 , both concentrations of NBP have significantly increased the viability of the SH-SY5Y cells ($P < 0.05$, Fig. 6). Under the influence of H_2O_2 , NBP at any concentration could still reduce the damage significantly ($P < 0.05$ or $P < 0.01$, Figs. 7, 8, 9), thus showing better cell survival.

Discussion

Our results illustrated that there was a significant increase of the blood vessel density after application of the drug, NBP, onto the CAM. Since the increased vessels around the carrier were of small sizes, they were probably newly formed vessels. The critical dosage for this effect was at 1 μM . At the same time, immunohistochemistry had shown an increase in reaction activities in VEGF, VEGF Receptor

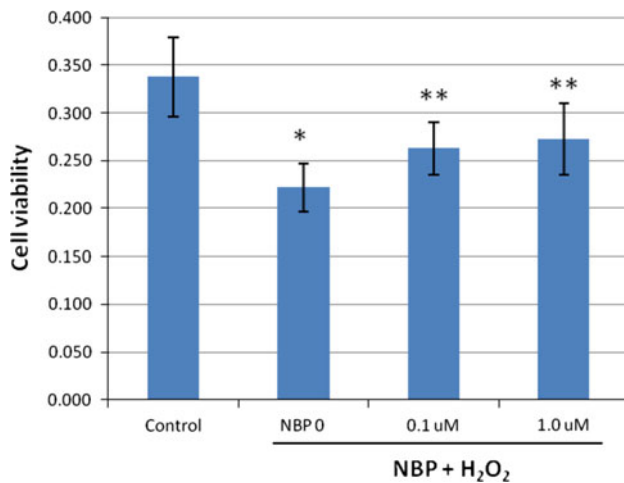


Fig. 8 The effects of simultaneously treating different concentrations of NBP and H_2O_2 to the viability of SH-SY5Y cells. Compared with control, * $P < 0.01$; compared with NBP 0, ** $P < 0.05$

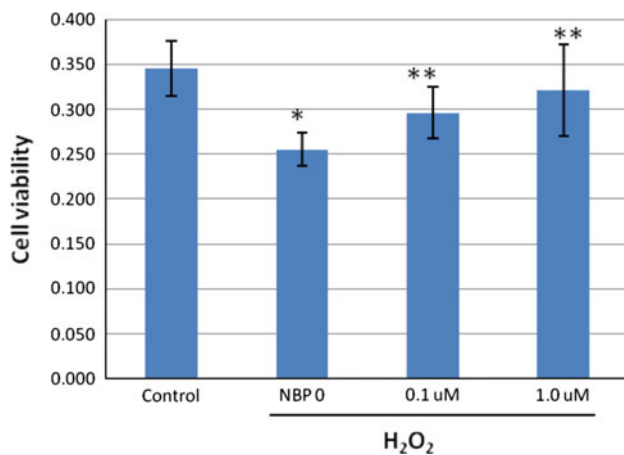


Fig. 9 The effects of pretreating different concentrations of NBP to H_2O_2 caused SH-SY5Y cell damage. Compared with control, * $P < 0.01$; compared with NBP 0, ** $P < 0.05$

2 and bFGF on the surface of the membrane and around the blood vessels, which suggested the possibility of increased angiogenesis. In fact, VEGF and bFGF had long been associated with angiogenesis [23–27], therefore it was highly likely that the increase in blood vessel density on the membrane was caused by the increased levels of VEGF, VEGF Receptor and bFGF induced by NBP. Apart from the increase in blood vessel density, Chong and Feng [15] have also reported an increase in blood flow upon the use of NBP in rats with induced subarachnoid hemorrhage. Furthermore, NBP could also protect cells from damages via other pathways, for example, increasing GSH in cells, suppressing mitochondria permeability [10] and interfering with proapoptotic signalling by downregulating phospho-JNK and p38 [12]. In the Parkinson animal model, NBP

had been shown to inhibit the production of ROS and protected dopaminergic neurons and terminals [11], it had also inhibited neuronal autophagy in the diabetic animal model [28]. It appeared that NBP did have multipotential in alleviating neurovascular and other neurological episodes. It would therefore be of worldwide interest to conduct more studies on this drug.

Formation of ROS is involved in the induction of neuronal cell death, for example after the addition of H_2O_2 in neuronal cultures [29–33]. The presence of these ROS has resulted in the initiation of apoptosis [29–33]. In the present work, H_2O_2 had also shown to induce cell death in the SH-SY5Y cell line as predicted, and the application of NBP had protected the cells from dying in all cases when NBP was pretreated, simultaneously treated or post-treated with H_2O_2 . The additional percentage in the survival of cells upon NBP treatment ranged from 10 to 25% and the best protective results were obtained upon pretreatment, i.e. NBP treatment prior to H_2O_2 treatment. Interestingly, the addition of NBP to normal cultures without H_2O_2 treatment demonstrated a higher level of cell survival, indicating the ability of NBP in the induction of neuronal growth and survival, even in normal cells. The induction of cell death by H_2O_2 treatment might be related to the damage on trophic factor [31], fluctuation of AKT, c-Jun N-terminal Kinase and caspase 3 [32], while protection from apoptosis might rely on the modulation of protein kinase C, ERK kinase 1/2 or phosphatidylinositol'3 pathway. This experiment showed that NBP indeed had a protective effect in preventing cell death in cell cultures.

In the in vivo rat model, bilateral ligation and transection were performed on the common carotid arteries, and damages were induced via production of global ischemia [28, 34]. Such global ischemic damage on the brain would affect the vascular supply of the hippocampus, which was essential to the learning and memory of the rats [17]. Learning and memory was often tested in rats via the Morris Water Maze test [17]. In this aspect, addition of NBP had shown to improve the performance of the rats that had bilaterally ligated common carotid arteries. This phenomenon appeared to demonstrate and again proved the neuronal protection ability of NBP against external insults. The advantage of NBP in this study actually protected both hypertensive and normotensive rats upon ischemia. Both in vitro (cell culture) and in vivo (rats) experiments had implied the role of NBP in protecting against damages. Therefore, further study to investigate the protective role of this drug on the nervous system is of clinical importance.

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Conflict of interest There is no conflict of interest.

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